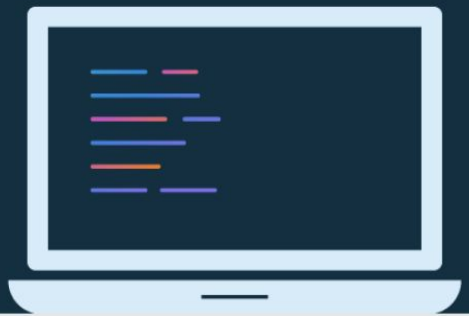


Lesson 7: Activity and fragment lifecycles



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About this lesson

Lesson 7: Activity and fragment lifecycles

- Activity lifecycle
- Logging
- Fragment lifecycle
- Lifecycle-aware components
- Tasks and back stack
- Summary

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Activity lifecycle

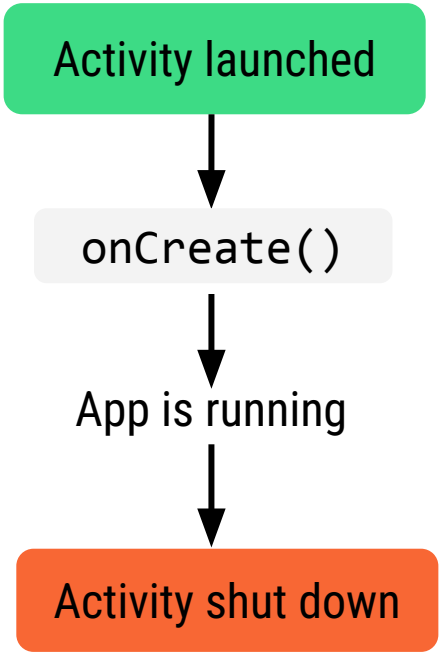
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Why it matters

- Preserve user data and state if:
  - User temporarily leaves app and then returns
  - User is interrupted (for example, a phone call)
  - User rotates device
- Avoid memory leaks and app crashes.

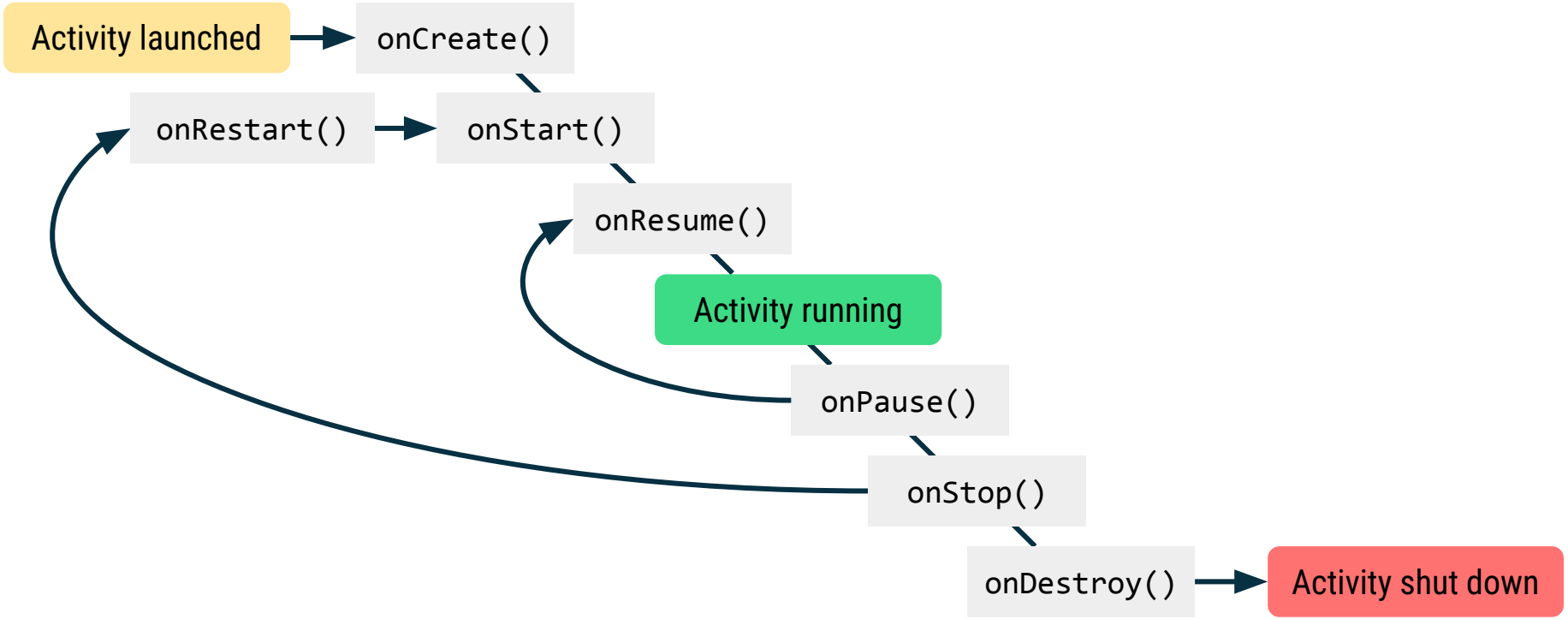
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Simplified activity lifecycle



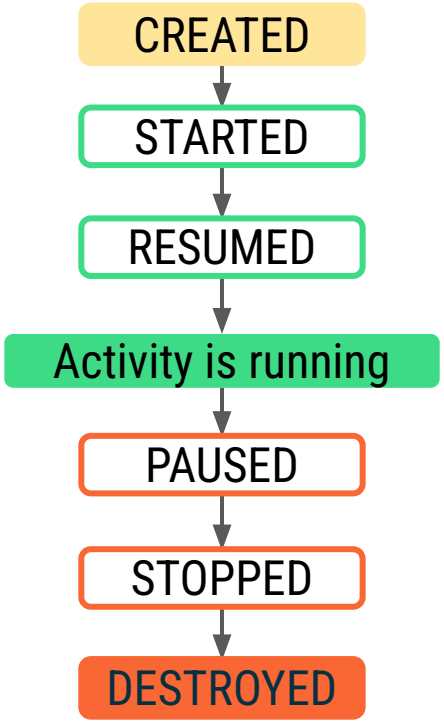
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Activity lifecycle



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Activity states



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onCreate()

- Activity is created and other initialization work occurs
- You must implement this callback
- Inflate activity UI and perform other app startup logic

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onStart()

- Activity becomes visible to the user
- Called after activity:
  - `onCreate()`

    or
  - `onRestart()` if activity was previously stopped

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onResume()

- Activity gains input focus:
  - User can interact with the activity
- Activity stays in resumed state until system triggers activity to be paused

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onPause()

- Activity has lost focus (not in foreground)
- Activity is still visible, but user is not actively interacting with it
- Counterpart to `onResume()`

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onStop()

- Activity is no longer visible to the user
- Release resources that aren't needed anymore
- Save any persistent state that the user is in the process of editing so they don't lose their work

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onDestroy()

- Activity is about to be destroyed, which can be caused by:
  - Activity has finished or been dismissed by the user
  - Configuration change
- Perform any final cleanup of resources.
- Don’t rely on this method to save user data (do that earlier)

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Summary of activity states

State	Callbacks	Description
Created	`onCreate()`	Activity is being initialized.
Started	`onStart()`	Activity is visible to the user.
Resumed	`onResume()`	Activity has input focus.
Paused	`onPause()`	Activity does not have input focus.
Stopped	`onStop()`	Activity is no longer visible.
Destroyed	`onDestroy()`	Activity is destroyed.

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Save state

User expects UI state to stay the same after a config change or if the app is terminated when in the background.

- Activity is destroyed and restarted, or app is terminated and activity is started.
- Store user data needed to reconstruct app and activity Lifecycle changes:
  - Use `Bundle` provided by `onSaveInstanceState()`.
  - `onCreate()` receives the `Bundle` as an argument when activity is created again.

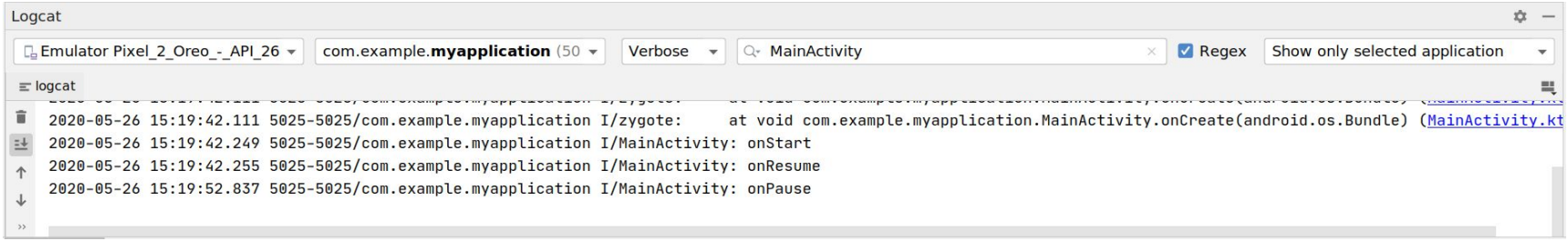
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Logging

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Logging in Android

- Monitor the flow of events or state of your app.
- Use the built-in `Log` class or third-party library.
- Example `Log` method call: `Log.d(TAG, "Message")`



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Write logs

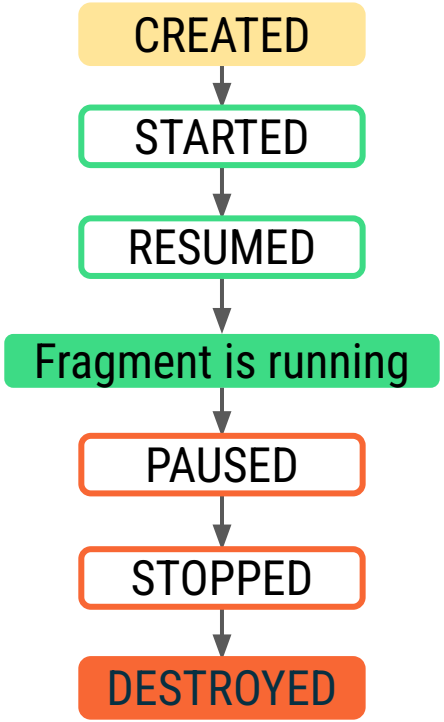
Priority level	Log method
Verbose	`Log.v(String, String)`
Debug	`Log.d(String, String)`
Info	`Log.i(String, String)`
Warning	`Log.w(String, String)`
Error	`Log.e(String, String)`

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Fragment lifecycle

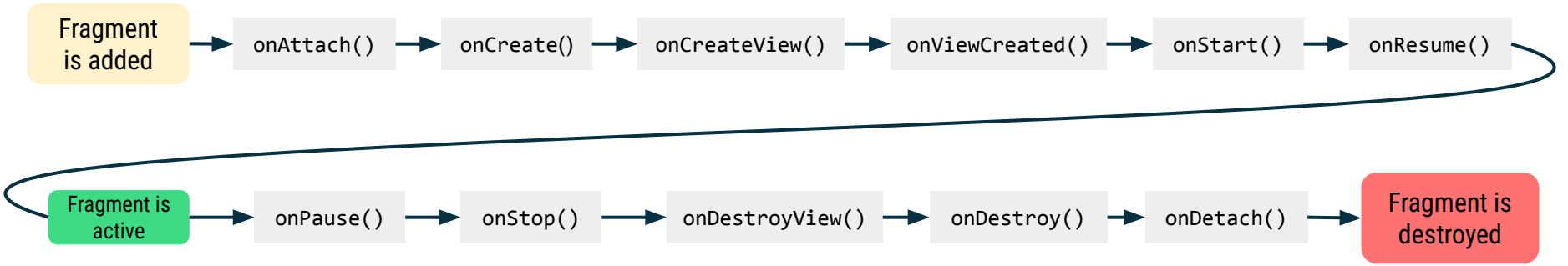
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Fragment states



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Fragment lifecycle diagram



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onAttach()

- Called when a fragment is attached to a context
- Immediately precedes `onCreate()`

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onCreateView()

- Called to create the view hierarchy associated with the fragment
- Inflate the fragment layout here and return the root view

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onViewCreated()

- Called when view hierarchy has already been created
- Perform any remaining initialization here (for example, restore state from `Bundle`)

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onDestroyView() and onDetach()

- `onDestroyView()` is called when view hierarchy of fragment is removed.
- `onDetach()` is called when fragment is no longer attached to the host.

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Summary of fragment states

State	Callbacks	Description
Initialized	`onAttach()`	Fragment is attached to host.
Created	`onCreate()`, `onCreateView()`, `onViewCreated()`	Fragment is created and layout is being initialized.
Started	`onStart()`	Fragment is started and visible.
Resumed	`onResume()`	Fragment has input focus.
Paused	`onPause()`	Fragment no longer has input focus.
Stopped	`onStop()`	Fragment is not visible.
Destroyed	`onDestroyView()`, `onDestroy()`, `onDetach()`	Fragment is removed from host.

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Save fragment state across config changes

Preserve UI state in fragments by storing state in `Bundle`:

- `onSaveInstanceState(outState: Bundle)`

Retrieve that data by receiving the Bundle in these fragment callbacks:

- `onCreate()`
- `onCreateView()`
- `onViewCreated()`

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Lifecycle-aware components

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Lifecycle-aware components

Adjust their behavior based on activity or fragment lifecycle

- Use the `androidx.lifecycle` library
- `Lifecycle` tracks the lifecycle state of an activity or fragment
  - Holds current lifecycle state
  - Dispatches lifecycle events (when there are state changes)

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LifecycleOwner

- Interface that says this class has a lifecycle
- Implementers must implement `getLifecycle()` method

  Examples: `Fragment` and `AppCompatActivity` are implementations of `LifecycleOwner`

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LifecycleObserver

Implement `LifecycleObserver` interface:

```
class MyObserver : LifecycleObserver {
    @OnLifecycleEvent(Lifecycle.Event.ON_RESUME)
    fun connectListener() {
        ...
    }
```

Add the observer to the lifecycle:

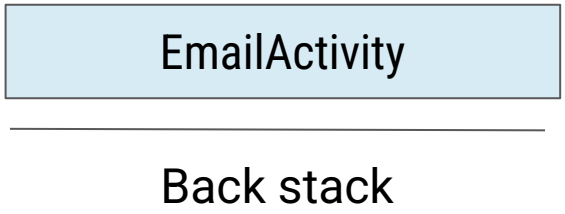
```
myLifecycleOwner.getLifecycle().addObserver(MyObserver())
```


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Tasks and back stack

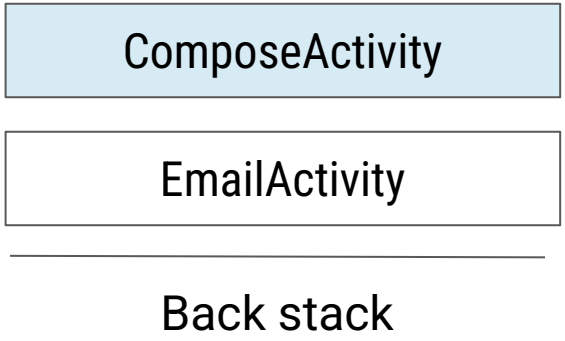
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Back stack of activities



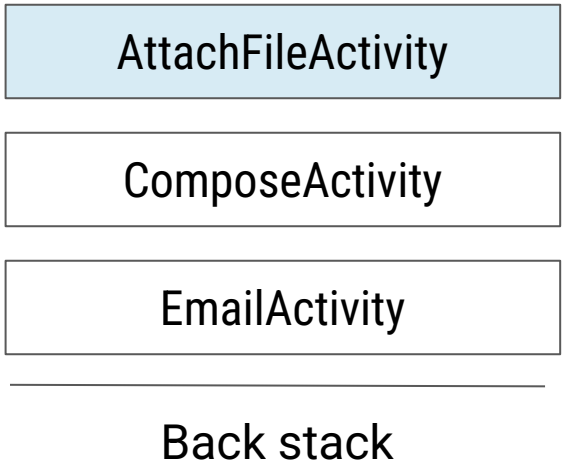
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Add to the back stack



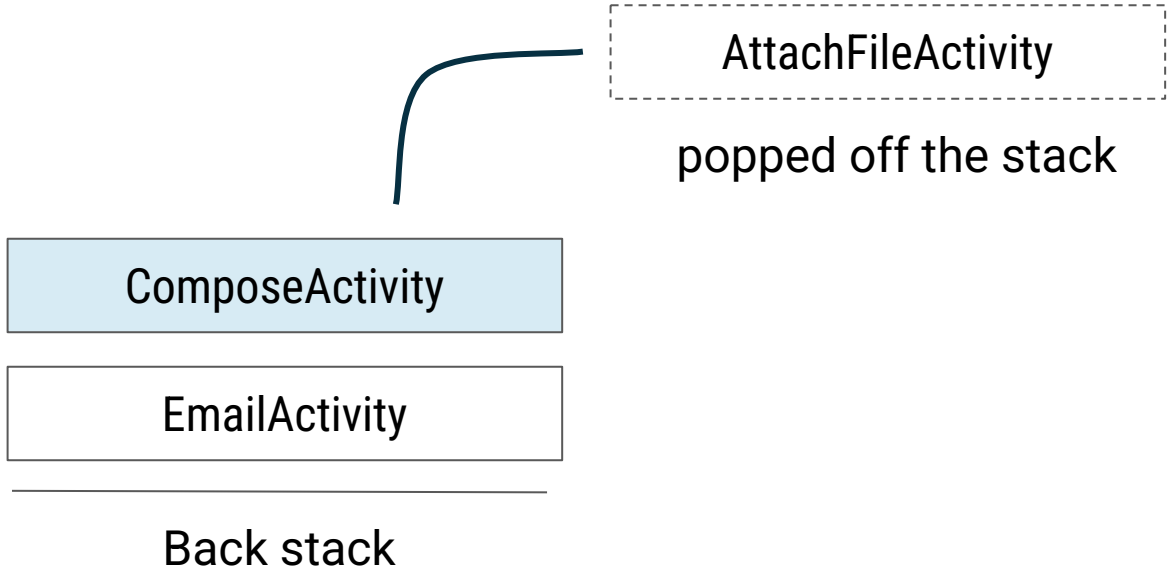
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Add to the back stack again



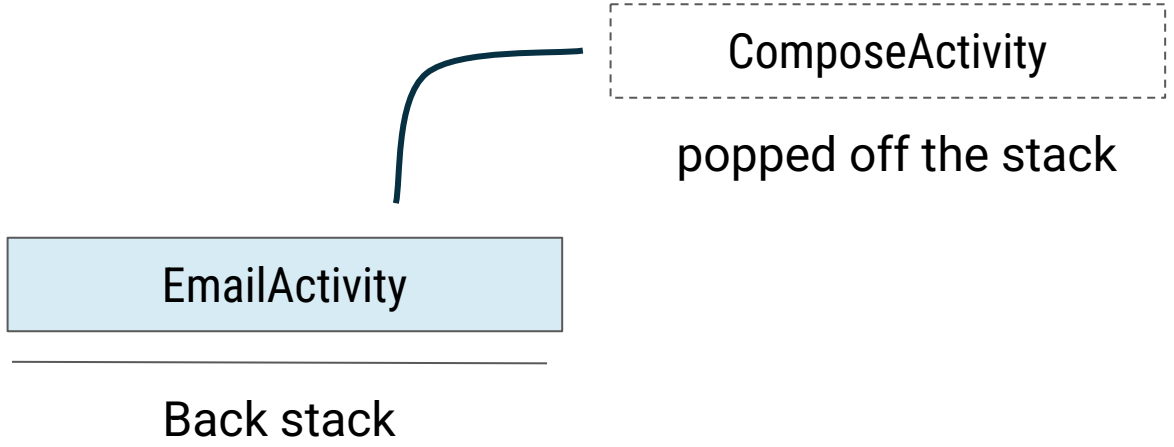
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Tap Back button



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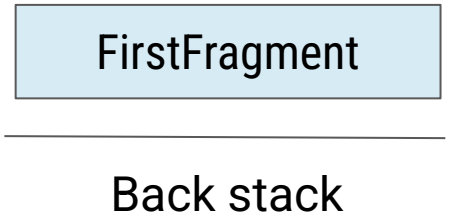
Tap Back button again



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First destination in the back stack

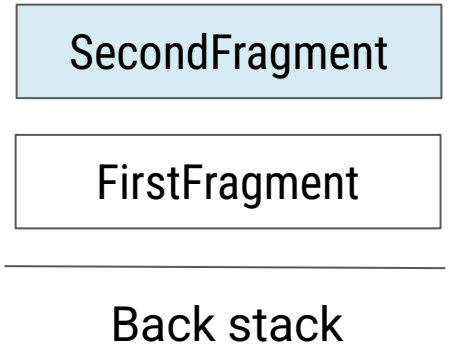




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Add a destination to the back stack

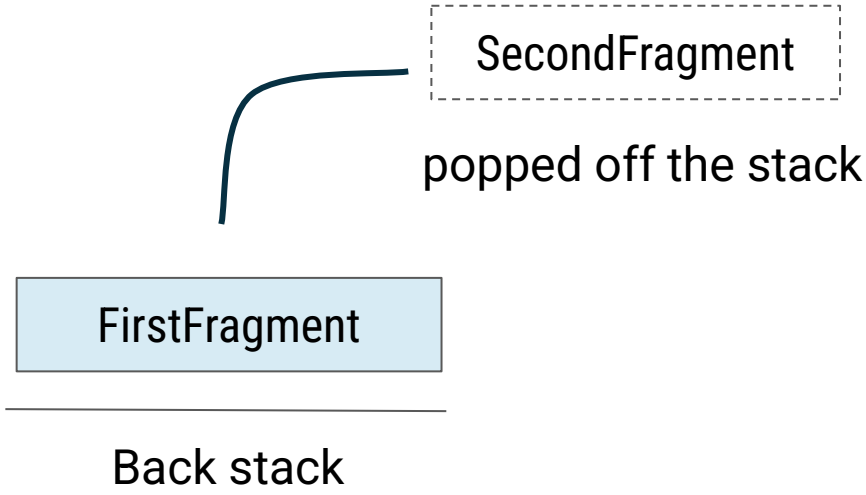




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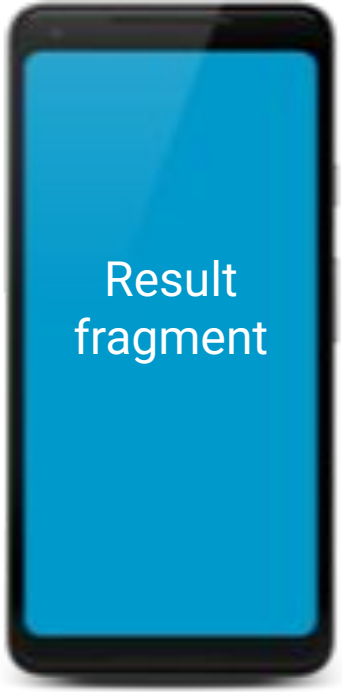
Tap Back button

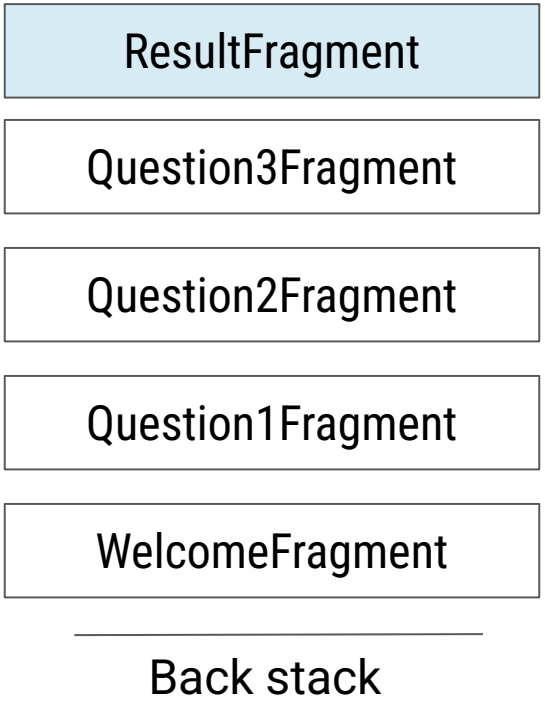




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Another back stack example



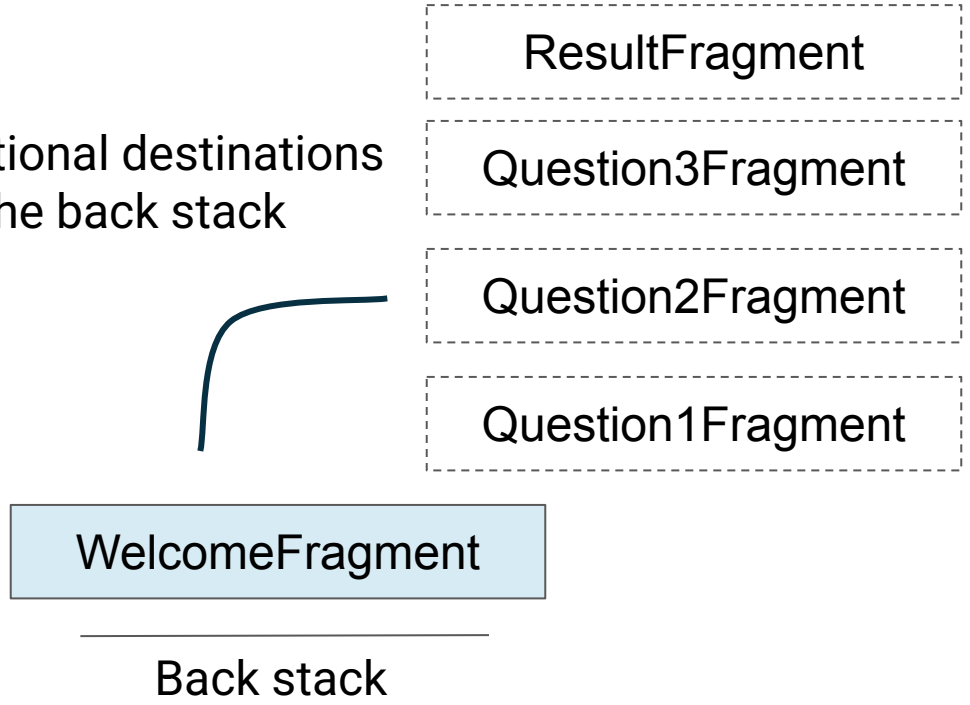


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Modify Back button behavior



pop additional destinations
off the back stack



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Summary

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Summary

In Lesson 7, you learned how to:

- Understand how an activity instance transitions through different lifecycle states as the user interacts with or leaves your app
- Reserve UI state across configuration changes using a `Bundle`
- Fragment lifecycle callback methods similar to activity, but with additions
- Use lifecycle-aware components help organize your app code
- Use default or custom back stack behavior
- Use logging to help debug and track the state of the app

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Learn more

- Understand the Activity Lifecycle
- Activity class
- Fragments guide and lifecycle
- Fragment class

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Pathway

Practice what you've learned by completing the pathway:

Lesson 7: Activity and Fragment Lifecycles



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